A REGULARIZATION OF THE THREE-BODY PROBLEM

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Abstract. Let $\mathbf{r}_1$, $\mathbf{r}_2$, $\mathbf{r}_3$ be arbitrary coordinates of the non-zero interacting mass-points m_1, m_2, m_3 and define the distances $R_1 = |\mathbf{r}_1 - \mathbf{r}_3|$, $R_2 = |\mathbf{r}_2 - \mathbf{r}_3|$, $R = |\mathbf{r}_1 - \mathbf{r}_2|$. An eight-dimensional regularization of the general three-body problem is given which is based on Kustaanheimo-Stiefel regularization of a single binary and possesses the properties:

(i) The equations of motion are regular for the two-body collisions $R_1 \to 0$ or $R_2 \to 0$.

(ii) Provided that $R \gtrsim R_1$ or $R \gtrsim R_2$, the equations of motion are numerically well behaved for close triple encounters.

Although the requirement $R \gtrsim \min(R_1, R_2)$ may involve occasional transformations to physical variables in order to re-label the particles, all integrations are performed in regularized variables. Numerical comparisons with the standard Kustaanheimo-Stiefel regularization show that the new method gives improved accuracy per integration step at no extra computing time for a variety of examples. In addition, time reversal tests indicate that critical triple encounters may now be studied with confidence.

The Hamiltonian formulation has been generalized to include the case of perturbed three-body motions and it is anticipated that this procedure will lead to further improvements of N-body calculations.

1. Introduction

According to Whittaker (1904), the general three-body problem is the most celebrated problem in dynamical astronomy. Considerable theoretical and numerical efforts have been devoted to the study of this fascinating subject but, for reasons of simplicity, most investigations have been concerned with the corresponding circular restricted problem. However, the numerical approach is more versatile, permitting arbitrary configurations to be considered. The rapid advance of high-speed computers has therefore increased the interest in the general three-body problem. At the same time, the numerical challenge has led to the introduction of many new techniques. In the first instance, classical perturbation methods which are based on the concept of dominant two-body motion have been used to calculate the orbits of planets and comets. Nevertheless, direct integration methods which are not restricted to small perturbations continue to enjoy considerable popularity because of their simplicity. On the other hand, the effectiveness of direct calculations is reduced during close approaches between the interacting bodies. Such events occur frequently in three-body systems; consequently, the accuracy of the solutions cannot be guaranteed in many cases. General considerations suggest that an ideal procedure should combine the greater flexibility of direct calculations with the advantage of using a perturbed two-body description, which gives regular resolutions in the neighbourhood of attracting centres.

The first example of regularization is due to Euler (1765) who employed a simple time transformation to eliminate the collision singularity of two particles moving on a

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straight line. In a classical paper, Levi-Civita (1903) showed that one of the singularities of the planar three-body problem can be removed by introducing new coordinates as well as the time transformation of Euler. The resulting equations of motion are regular if only one distance tends to zero, and the corresponding solutions are then well behaved. However, the principle of Levi-Civita regularization is based on complex variables, which cannot be extended to three dimensions. For a long time this fundamental stumbling block prevented further progress, until Kustaanheimo and Stiefel (1965), by an elegant four-dimensional generalization of the Levi-Civita procedure, succeeded in regularizing the general two-body problem with arbitrary external perturbations. This remarkable achievement, subsequently referred to as standard KS regularization, has stimulated further significant advances and has also done much to popularize the principles of regularization. It is indicative of the numerical effectiveness that the regularized solutions of any dominant two-body motion are improved with respect to direct integration methods.

The history of three-body regularization dates back to the celebrated works of Poincaré (1907) and Sundman (1912). In fact, Sundman solved the general problem in principle, excepting the case of triple collision, by introducing two time transformations. Unfortunately, the solutions are specified by infinite power series which do not reveal the behaviour of the motions. It is also doubtful whether the three-body regularization proposed by Lemaître (1955) is practically useful in the general case. Consequently, the introduction of the KS theory can properly be said to have set the stage for a renewed attack on the dynamical summit represented by the general three-body problem. This challenge divides naturally into two parts:

(i) Regularization of all two-body collisions by one 'global' transformation.

(ii) Improved treatment of close triple encounters.

Examples of transformations satisfying the first requirement in the planar restricted three-body problem are due to Thiele (1896), Burrau (1906), Birkhoff (1915) and Lemaître (1955). An equivalent three-dimensional formulation of Birkhoff's regularization has been given by Stiefel and Waldvogel (1965); in all cases the body of infinitesimal mass can have repeated collisions with either of the two primaries without change of variables. It may be noted that, until quite recently, alternative methods satisfying the second requirement were not available. A new milestone was therefore reached when Waldvogel (1972) obtained a global regularization of the planar three-body problem with arbitrary masses. The corresponding equations of motion are symmetrical with respect to the individual mass-points and have the desirable property of satisfying both the requirements above.

Although current regularization theory is still not able to deal with the most general problem in a practical way, recent advances have been encouraging and permit new avenues to be explored. The present approach is based on familiar ideas of Hamiltonian dynamics in the extended phase space, and employs the six centre of mass integrals to reduce the order of the differential equations which describe the particle motions. In the following derivation we rely heavily on the classical canonical treatment of Whittaker (1904), as well as the modern discourse on regularization by

Szebehely (1967). The actual regularization is achieved by introducing two KS transformations. This permits one of the particles to have repeated collisions with the other two, whose relative motion is described by equations which are still singular but numerically well behaved.

2. Reduction of the Order

Following Whittaker (1904), let a general three-body configuration be described by the particle masses m_1, m_2, m_3, with respective coordinates (q'_1, q'_2, q'_3), (q'_4, q'_5, q'_6), (q'_7, q'_8, q'_9) and corresponding momenta (p'_1, p'_2, p'_3), (p'_4, p'_5, p'_6), (p'_7, p'_8, p'_9). The geometry of the problem is defined by Figure 1 which also shows the three distances

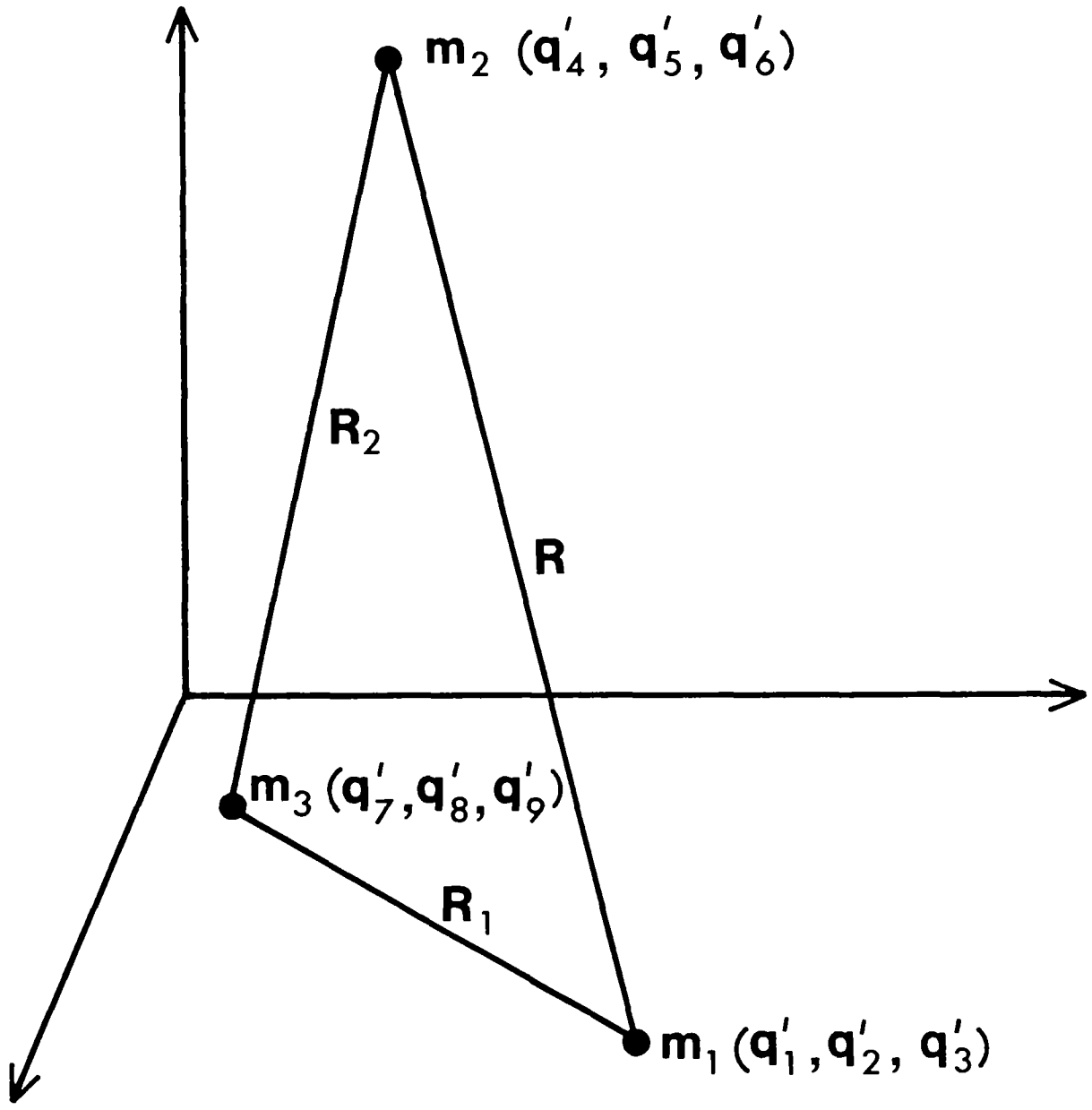


Fig. 1. The geometry of a general three-body configuration.

R_1, R_2, R. If the constant of gravity is taken to be unity, this system has the Hamiltonian

$$H' = \sum_{r=1}^{9} \frac{p_r'^2}{2m_k} - \frac{m_1 m_3}{R_1} - \frac{m_2 m_3}{R_2} - \frac{m_1 m_2}{R}, \tag{1}$$

where k denotes the integer part of $\frac{1}{3}(r+2)$ and

$$\begin{aligned} R_1 &= [(q'_1 - q'_7)^2 + (q'_2 - q'_8)^2 + (q'_3 - q'_9)^2]^{1/2}, \\ R_2 &= [(q'_4 - q'_7)^2 + (q'_5 - q'_8)^2 + (q'_6 - q'_9)^2]^{1/2}, \\ R &= [(q'_1 - q'_4)^2 + (q'_2 - q'_5)^2 + (q'_3 - q'_6)^2]^{1/2}. \end{aligned} \tag{2}$$

The equations of motion with respect to the time t are

$$\frac{\mathrm{d}q'_r}{\mathrm{d}t} = \frac{\partial H'}{\partial p'_r}, \qquad \frac{\mathrm{d}p'_r}{\mathrm{d}t} = -\frac{\partial H'}{\partial q'_r}. \qquad (r = 1, 2, \ldots, 9) \tag{3}$$

This system of 18 first-order differential equations may be reduced to a system of six equations by means of the ten integrals of motion as well as elimination of the nodes and the time. However, in the present formulation the order of the problem is reduced by six only, using the centre of mass integrals. The variables q'_r, p'_r are first transformed to new variables q_r, p_r by the canonical contact transformations

$$q'_r = \frac{\partial W}{\partial p'_r}, \qquad p_r = \frac{\partial W}{\partial q_r}, \qquad (r = 1, 2, \ldots, 9) \tag{4}$$

where the generating function W is given by

$$W = \sum_{r=1}^{6} p'_r q_r + \sum_{r=1}^{3} (p'_r + p'_{r+3} + p'_{r+6})\, q_{r+6}. \tag{5}$$

The variables q_r can be obtained by inverting the first of Equations (4). It can be verified that the new variables have the following simple physical interpretation: (q_1, q_2, q_3) and (q_4, q_5, q_6) are the relative coordinates of m_1 and m_2 with respect to m_3, (q_7, q_8, q_9) are the coordinates of m_3, (p_1, p_2, p_3) and (p_4, p_5, p_6) are the momenta of m_1 and m_2, respectively, whereas (p_7, p_8, p_9) are the momenta of the centre of mass.

Substituting the new variables for the old in Equation (1), it can be seen that q_7, q_8, q_9 are absent from the Hamiltonian and are therefore ignorable coordinates. Consequently, the corresponding momenta are then integrals of the motion. Without loss of generality we may adopt

$$p_r = 0, \qquad (r = 7, 8, 9) \tag{6}$$

which implies that the original coordinates and momenta are expressed with respect to the centre of mass. Hence the Hamiltonian simplifies to

$$H = \sum_{r=1}^{6} \frac{1}{2\mu_{k3}} p_r^2 + \frac{1}{m_3} \sum_{r=1}^{3} p_r p_{r+3} - \frac{m_1 m_3}{R_1} - \frac{m_2 m_3}{R_2} - \frac{m_1 m_2}{R}, \tag{7}$$

with the reduced mass $\mu_{k3} = m_k m_3/(m_k + m_3)$ and the particle separations

$$\begin{aligned} R_1 &= (q_1^2 + q_2^2 + q_3^2)^{1/2}, \\ R_2 &= (q_4^2 + q_5^2 + q_6^2)^{1/2}, \\ R &= [(q_1 - q_4)^2 + (q_2 - q_5)^2 + (q_3 - q_6)^2]^{1/2}. \end{aligned} \tag{8}$$

The equations of motion now become

$$\frac{\mathrm{d}q_r}{\mathrm{d}t} = \frac{\partial H}{\partial p_r}, \qquad \frac{\mathrm{d}p_r}{\mathrm{d}t} = -\frac{\partial H}{\partial q_r}. \qquad (r = 1, 2, \ldots, 6) \tag{9}$$

The employment of the six integrals of the centre of mass motion therefore gives rise to an equivalent set of 12 first-order equations. However, these equations still contain the original singularities which may be removed by introducing regularized variables.

3. Formulation in Eight Dimensions

Because of fundamental difficulties originally clarified by Hopf (1931) and Hurwitz (1933), it is impossible to generalize the Levi-Civita transformation to an equivalent set of three-dimensional variables. This problem was eventually surmounted when Kustaanheimo and Stiefel introduced their four-dimensional representation of two-body motion with three degrees of freedom. The equations of motion (9) describe a system with six degrees of freedom, hence it is natural to perform regularization in eight dimensions by increasing the set of physical variables accordingly.

First we re-label some of the coordinates and momenta by

$$q_r \to q_{r+1}, \qquad p_r \to p_{r+1}, \qquad (r = 4, 5, 6) \tag{10}$$

leaving all variables with $r \leqslant 3$ as before. The corresponding Hamiltonian no longer depends on quantities with subscripts $r=4$ or $r=8$; consequently, we introduce additional variables defined by

$$q_4 \equiv 0, \qquad q_8 \equiv 0, \qquad p_4 \equiv 0, \qquad p_8 \equiv 0. \tag{11}$$

The new Hamiltonian takes the form

$$H = \sum_{r=1}^{8} \frac{1}{2\mu_{k3}} p_r^2 + \frac{1}{m_3} \sum_{r=1}^{4} p_r p_{r+4} - \frac{m_1 m_3}{R_1} - \frac{m_2 m_3}{R_2} - \frac{m_1 m_2}{R}, \tag{12}$$

where k is now the integer part of $\frac{1}{4}(r+3)$ and

$$R_2 = (q_5^2 + q_6^2 + q_7^2)^{1/2}. \tag{13}$$

If the additional variables are canonical, we have the equations of motion

$$\frac{\mathrm{d}q_r}{\mathrm{d}t} = \frac{\partial H}{\partial p_r}, \qquad \frac{\mathrm{d}p_r}{\mathrm{d}t} = -\frac{\partial H}{\partial q_r}. \qquad (r = 1, 2, \ldots, 8) \tag{14}$$

It can be readily verified that Equation (14) is consistent with the identity conditions (11).

Next we introduce a new set of canonical variables in eight dimensions, denoted by $\mathbf{Q}=(Q_1, Q_2, \ldots, Q_8)$, $\mathbf{P}=(P_1, P_2, \ldots, P_8)$. According to the theory of canonical contact transformations, if we choose the generating function

$$W = W(\mathbf{p}, \mathbf{Q}), \tag{15}$$

then

$$q_r = \frac{\partial W}{\partial p_r}, \qquad P_r = \frac{\partial W}{\partial Q_r}. \qquad (r = 1, 2, \ldots, 8) \tag{16}$$

In order to avoid complicated expressions, we follow Szebehely (1967) and adopt a separable generating function of the form

$$W = \sum_{r=1}^{8} p_r f_r , \tag{17}$$

with the conditions

$$\begin{aligned} f_r &= f_r(Q_1, Q_2, Q_3, Q_4), \qquad (r \leqslant 4) \\ f_r &= f_r(Q_5, Q_6, Q_7, Q_8). \qquad (r > 4) \end{aligned} \tag{18}$$

Substitution into the first of Equations (16) then yields the simple result

$$q_r = f_r , \qquad (r = 1, 2, \ldots, 8), \tag{19}$$

where the first four coordinates q_r only depend on the variables Q_1, Q_2, Q_3, Q_4, etc. The corresponding transformation equation for the P-variables becomes

$$P_r = \sum_{j=1}^{8} p_j \frac{\partial f_j}{\partial Q_r} . \qquad (r = 1, 2, \ldots, 8) \tag{20}$$

It is convenient to write Equation (20) in the vector form

$$\mathbf{P} = \mathbf{A}\mathbf{p} , \tag{21}$$

where

$$\mathbf{A} = \begin{bmatrix} \frac{\partial f_1}{\partial Q_1} & \frac{\partial f_2}{\partial Q_1} & \frac{\partial f_3}{\partial Q_1} & \frac{\partial f_4}{\partial Q_1} & 0 & 0 & 0 & 0 \\ \frac{\partial f_1}{\partial Q_2} & \frac{\partial f_2}{\partial Q_2} & \frac{\partial f_3}{\partial Q_2} & \frac{\partial f_4}{\partial Q_2} & 0 & 0 & 0 & 0 \\ \frac{\partial f_1}{\partial Q_3} & \frac{\partial f_2}{\partial Q_3} & \frac{\partial f_3}{\partial Q_3} & \frac{\partial f_4}{\partial Q_3} & 0 & 0 & 0 & 0 \\ \frac{\partial f_1}{\partial Q_4} & \frac{\partial f_2}{\partial Q_4} & \frac{\partial f_3}{\partial Q_4} & \frac{\partial f_4}{\partial Q_4} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\partial f_5}{\partial Q_5} & \frac{\partial f_6}{\partial Q_5} & \frac{\partial f_7}{\partial Q_5} & \frac{\partial f_8}{\partial Q_5} \\ 0 & 0 & 0 & 0 & \frac{\partial f_5}{\partial Q_6} & \frac{\partial f_6}{\partial Q_6} & \frac{\partial f_7}{\partial Q_6} & \frac{\partial f_8}{\partial Q_6} \\ 0 & 0 & 0 & 0 & \frac{\partial f_5}{\partial Q_7} & \frac{\partial f_6}{\partial Q_7} & \frac{\partial f_7}{\partial Q_7} & \frac{\partial f_8}{\partial Q_7} \\ 0 & 0 & 0 & 0 & \frac{\partial f_5}{\partial Q_8} & \frac{\partial f_6}{\partial Q_8} & \frac{\partial f_7}{\partial Q_8} & \frac{\partial f_8}{\partial Q_8} \end{bmatrix}, \tag{22}$$

and $\mathbf{p}$ is the column vector with components p_j. Hence the adopted generating function (17) gives rise to a transformation matrix $\mathbf{A}$ which can be sub-divided into two matrices of order four. This property permits Equation (21) to be split into the two component equations

$$\mathbf{P}_k = \mathbf{A}_k \mathbf{p}_k , \qquad (k = 1, 2) \tag{23}$$

where $\mathbf{A}_1$ and $\mathbf{A}_2$ constitute the upper and lower 4×4 sub-matrix, respectively, and $\mathbf{P}_1 = (P_1, P_2, P_3, P_4)$, etc. The corresponding momenta are obtained by the inversion

$$\mathbf{p}_k = \mathbf{A}_k^{-1} \mathbf{P}_k . \qquad (k = 1, 2) \tag{24}$$

Let us assume that the matrix $\mathbf{A}_k$ satisfies the orthogonality condition

$$\mathbf{A}_k \mathbf{A}_k^T = D_k \mathbf{I} , \qquad (k = 1, 2) \tag{25}$$

where $\mathbf{A}_k^T$ is the transpose of $\mathbf{A}_k$, D_k is a scalar and $\mathbf{I}$ is the identity matrix. Substitution for $\mathbf{A}_k^{-1}$ in Equation (24) then gives

$$\mathbf{p}_k = \mathbf{A}_k^T \mathbf{P}_k / D_k . \qquad (k = 1, 2) \tag{26}$$

The definition of the column vector $\mathbf{p}_k$ implies that

$$\mathbf{p}_k^2 = \mathbf{p}_k^T \mathbf{p}_k . \qquad (k = 1, 2) \tag{27}$$

Applying the rule for the transpose of matrix products, we obtain the relation

$$\mathbf{p}_k^2 = \mathbf{P}_k^2 / D_k . \qquad (k = 1, 2) \tag{28}$$

The transformed Hamiltonian, subsequently denoted by $\tilde{H}$, is equal to the Hamiltonian of Equation (12) since the generating function (15) does not contain the time explicitly. Substitution for the momentum vectors (26) and the scalars (28) in Equation (12) yields the new Hamiltonian

$$\tilde{H} = \sum_{k=1}^{2} \frac{1}{2\mu_{k3}} \frac{\mathbf{P}_k^2}{D_k} + \frac{\mathbf{P}_1^T \mathbf{A}_1 \mathbf{A}_2^T \mathbf{P}_2}{m_3 D_1 D_2} - \frac{m_1 m_3}{R_1} - \frac{m_2 m_3}{R_2} - \frac{m_1 m_2}{R} , \tag{29}$$

where R_1, R_2, R are now expressed in terms of transformed coordinates, with

$$\begin{aligned} R_1 &= (f_1^2 + f_2^2 + f_3^2)^{1/2} , \\ R_2 &= (f_5^2 + f_6^2 + f_7^2)^{1/2} . \end{aligned} \tag{30}$$

Because of the identity conditions (11), the scalar product of Equation (29) contains only three non-zero terms as in the original expression (7).

Following the notation of Szebehely (1967), we introduce the Hamiltonian function Γ in the extended phase space, as well as a new independent variable w, and adopt $Q_9 = t$, $P_9 = -H$ as the additional canonical variables. It can be shown that if

$$\Gamma = P_9 + \tilde{H} , \tag{31}$$

the corresponding equations of motion become

$$\frac{\mathrm{d}Q_r}{\mathrm{d}w} = \frac{\partial \Gamma}{\partial P_r} , \qquad \frac{\mathrm{d}P_r}{\mathrm{d}w} = -\frac{\partial \Gamma}{\partial Q_r} , \qquad (r = 1, 2, \ldots, 8) \tag{32}$$

$$\frac{\mathrm{d}Q_9}{\mathrm{d}w} = 1 , \qquad \frac{\mathrm{d}P_9}{\mathrm{d}w} = 0 . \tag{33}$$

The equation for Q_9 implies that $w = t + \text{const.}$, hence $\mathrm{d}w = \mathrm{d}t$. On the other hand, the derivative of P_9 is equal to zero because Γ does not depend explicitly on the time; this result also follows from the assumption that $P_9 = -H = \text{const.}$

4. Regularization Procedure

Although the Hamiltonian (31) is still not regular, it takes a simple form which enables the final regularization to be performed. Any practically useful regularization formulation consists of introducing a coordinate transformation or equivalent regular two-body elements, as well as a new time. Before stating the former explicitly, we write the time transformation

$$\mathrm{d}t = g\,\mathrm{d}\tau\,, \tag{34}$$

where $g = g(Q_1, Q_2, \ldots, Q_8)$ and τ is the new independent variable. If Γ is replaced by

$$\Gamma^* = g\Gamma\,, \tag{35}$$

the canonical form of the equations of motion (32) are retained, giving

$$\frac{\mathrm{d}Q_r}{\mathrm{d}\tau} = \frac{\partial \Gamma^*}{\partial P_r}\,, \qquad \frac{\mathrm{d}P_r}{\mathrm{d}\tau} = -\frac{\partial \Gamma^*}{\partial Q_r}\,. \qquad (r = 1, 2, \ldots, 8) \tag{36}$$

We note that the condition† $P_9 + \tilde{H} = 0$ leads to the simple form of the final equations of motion for P_r, whereas the functional dependence of g on the coordinates only does not introduce further complications in the equations for Q_r. In order to regularize the part of the Hamiltonian associated with the kinetic energy, we choose

$$g = \lambda D_1 D_2\,, \tag{37}$$

where λ is an adjustable constant. After substitution, it may be seen that the singularities in Equation (35) are associated with the terms $D_1 D_2/R_1$, $D_1 D_2/R_2$ and $D_1 D_2/R$. Consequently, two of these singularities can be removed by adopting

$$D_k = \lambda_k R_k\,, \qquad (k = 1, 2) \tag{38}$$

where λ_k is a constant. The assumption (25) then takes the form

$$\mathbf{A}_k \mathbf{A}_k^T = \lambda_k R_k \mathbf{I}\,. \qquad (k = 1, 2) \tag{39}$$

This relation is satisfied by the KS transformations (Kustaanheimo and Stiefel, 1965)

$$\begin{aligned}
f_1 &= Q_1^2 - Q_2^2 - Q_3^2 + Q_4^2\,,\\
f_2 &= 2(Q_1 Q_2 - Q_3 Q_4)\,,\\
f_3 &= 2(Q_1 Q_3 + Q_2 Q_4)\,,\\
f_4 &= 0\,,\\
f_5 &= Q_5^2 - Q_6^2 - Q_7^2 + Q_8^2\,,\\
f_6 &= 2(Q_5 Q_6 - Q_7 Q_8)\,,\\
f_7 &= 2(Q_5 Q_7 + Q_6 Q_8)\,,\\
f_8 &= 0\,.
\end{aligned} \tag{40}$$

† This procedure is equivalent to the introduction of a control term in the equations of motion (Baumgarte and Stiefel, 1974).

These transformations have the well-known properties

$$\sum_{r=1}^{4} Q_r^2 = R_1\,, \qquad \sum_{r=5}^{8} Q_r^2 = R_2\,. \tag{41}$$

Furthermore, Equations (11) and (23) imply two bilinear relations of the type which appears in the standard KS theory, once the matrices $\mathbf{A}_k$ are completely specified.

The fourth column of the matrices $\mathbf{A}_k$ must be such that Equation (39) is satisfied; hence it can be constructed directly if required. However, it is sufficient for our purpose to write the orthogonality condition for $r=4$ and $r=8$, which in the former case becomes

$$\sum_{i=1}^{4} \frac{\partial f_4}{\partial Q_i}\frac{\partial f_j}{\partial Q_i} = \lambda_1 R_1 \delta_{4j}\,, \tag{42}$$

where δ_{ij} is the Kronecker symbol. The numerical constant is obtained by applying the analogue of Equation (42) to any other non-zero component, giving $\lambda_k = 4$. Hence Equation (37) is simplified by the choice $\lambda = \frac{1}{16}$, which leads to the time transformation

$$\mathrm{d}t = R_1 R_2\, \mathrm{d}\tau\,. \tag{43}$$

Consequently, the transformed Hamiltonian function takes the final form

$$\Gamma^* = R_1 R_2 P_9 + \sum_{k=1}^{2} \frac{1}{8\mu_{k3}} R_l \mathbf{P}_k^2 + \frac{1}{16 m_3}\mathbf{P}_1^T \mathbf{A}_1 \mathbf{A}_2^T \mathbf{P}_2 - \\ - m_1 m_3 R_2 - m_2 m_3 R_1 - m_1 m_2 R_1 R_2 / R\,, \tag{44}$$

where l is the integer part of $(k+1)/k$.

The relevant equations of motion may now be obtained by differentiation of Γ^* with respect to the canonical variables according to Equations (36). These equations are regular for $R_1 \to 0$ or $R_2 \to 0$, since Equation (28) implies that $|\mathbf{P}_k| \to$ const. as $R_k \to 0$, hence the mass-point m_3 can have collisions or close encounters with either m_1 or m_2 without causing numerical difficulties. Although the case $R \to 0$ is still singular, collisions between m_1 and m_2 may be studied by introducing relative coordinates with respect to either of these mass-points, rather than with respect to m_3. The calculation of successive collisions or close encounters between any pair of particles may therefore be carried out by a re-labelling procedure. In addition, the form of the last term of Equation (44) suggests that close triple encounters may also be integrated numerically, provided that $R \gtrsim \min(R_1, R_2)$. This requirement can be satisfied by an appropriate choice of reference body, in which case the critical term $R_1 R_2 / R$ is numerically well behaved even for small values of R. Indeed, it is mainly this latter property which distinguishes the present formulation from previous regularizations of three-dimensional problems.

The regularization method developed above is completely analogous to the KS

theory which also introduces transformations of both the coordinates and the time. The necessity of this approach may be clarified by the following remarks. In the first instance, removal of the singularity from the Hamiltonian does not necessarily imply the absence of singular terms in the equations of motion. Thus if the Hamiltonian is regular but contains terms of the type R_k^α, with $\alpha < 2$, the resulting equations of motion are not regular. Apart from the standard procedure of introducing new dependent and independent variables, singularities may also be removed from the original equations of motion by the time transformation $\mathrm{d}t = R_1^3 R_2^3 \, \mathrm{d}\tau$. However, actual regularization is not achieved in the second case because the new time becomes infinite at two-body collisions.

5. Transformations

This section describes the complete procedure for obtaining regularized quantities from the physical variables q_r', p_r' and vice versa. First define relative coordinates and re-define the momenta by

$$\begin{aligned} \mathbf{q}_k &= \mathbf{q}_k' - \mathbf{q}_3', \qquad (k = 1, 2) \\ \mathbf{p}_k &= \mathbf{p}_k', \qquad (k = 1, 2) \end{aligned} \tag{45}$$

where $\mathbf{q}_1 = (q_1, q_2, q_3)$, etc. Next expand the variables to eight dimensions by the re-labelling

$$q_r \to q_{r+1}, \qquad p_r \to p_{r+1}, \qquad (r = 4, 5, 6) \tag{46}$$

and the initialization of redundant variables,

$$q_4 \equiv 0, \qquad q_8 \equiv 0, \qquad p_4 \equiv 0, \qquad p_8 \equiv 0. \tag{47}$$

Regularized coordinates are now introduced by KS transformations. If $q_1 \geqslant 0$, the solution of Equation (40) for the first four new coordinates is obtained from

$$\begin{aligned} Q_1 &= [\tfrac{1}{2}(|\mathbf{q}_1| + q_1)]^{1/2}, \\ Q_2 &= q_2/2Q_1, \\ Q_3 &= q_3/2Q_1, \\ Q_4 &= 0, \end{aligned} \tag{48}$$

whereas if $q_1 < 0$, the appropriate choice is

$$\begin{aligned} Q_2 &= [\tfrac{1}{2}(|\mathbf{q}_1| - q_1)]^{1/2}, \\ Q_1 &= q_2/2Q_2, \\ Q_3 &= 0, \\ Q_4 &= q_3/2Q_2. \end{aligned} \tag{49}$$

Corresponding transformations for Q_5, Q_6, Q_7, Q_8 in terms of the physical variables q_5, q_6, q_7 are written in a similar way, replacing Q_1 by Q_5, etc. The regularized momenta are then obtained by means of

$$\mathbf{P}_k = \mathbf{A}_k \mathbf{p}_k. \qquad (k = 1, 2) \tag{50}$$

Because of the presence of the redundant variables (47), the numerical procedure only requires the 4×3 sub-matrix of $\mathbf{A}_k$, which for $k=1$ takes the explicit form

$$\mathbf{A}_1 = 2 \begin{bmatrix} Q_1 & Q_2 & Q_3 \\ -Q_2 & Q_1 & Q_4 \\ -Q_3 & -Q_4 & Q_1 \\ Q_4 & -Q_3 & Q_2 \end{bmatrix}. \tag{51}$$

We note that, apart from the factor two, this matrix is identical to the transpose of the generalized Levi-Civita matrix used in numerical applications of the standard KS regularization (Aarseth, 1971).

Inverse transformations are required to obtain the physical variables. Using matrix notation, the relative coordinates can be written as

$$\mathbf{q}_k = \tfrac{1}{2}\mathbf{A}_k^T \mathbf{Q}_k, \qquad (k = 1, 2) \tag{52}$$

where $\mathbf{Q}_1 = (Q_1, Q_2, Q_3, Q_4)$, etc. Transformation of the regularized momenta proceed according to

$$\mathbf{p}_k = \mathbf{A}_k^T \mathbf{P}_k / 4R_k. \qquad (k = 1, 2) \tag{53}$$

In order to recover the coordinates and momenta in the original six dimensions, variables with $r \geqslant 4$ are first re-labelled by

$$q_{r+1} \to q_r, \qquad p_{r+1} \to p_r. \qquad (r = 4, 5, 6) \tag{54}$$

Finally, the desired physical coordinates and momenta, expressed with respect to the centre of mass, are acquired from the expressions

$$\begin{aligned} \mathbf{q}_3' &= -\sum_{k=1}^{2} m_k \mathbf{q}_k / \sum_{k=1}^{3} m_k, \\ \mathbf{q}_k' &= \dot{\mathbf{q}}_3' + \mathbf{q}_k, \qquad (k = 1, 2) \\ \mathbf{p}_k' &= \mathbf{p}_k, \qquad (k = 1, 2) \\ \mathbf{p}_3' &= -(\mathbf{p}_1 + \mathbf{p}_2). \end{aligned} \tag{55}$$

6. The Perturbed Three-Body Problem

In view of the effectiveness of the KS regularization for studying close encounters in the N-body problem (Aarseth, 1971), it is natural to seek an equivalent derivation for perturbed three-body motions. Although the unperturbed treatment developed above requires only minor modifications in order to achieve this aim, it is convenient to introduce some new notation.

Consider a system of N particles with individual coordinates $\mathbf{r}_i$, velocities $\mathbf{v}_i$ and masses m_i. Without any loss of generality we assume as before that the three mass-points m_1, m_2, m_3 with global coordinates $\mathbf{r}_1, \mathbf{r}_2, \mathbf{r}_3$ form a sub-system to be regularized. Consequently, the Hamiltonian for the whole system may be written in the form

$$H_N = H_3 + \mathscr{R}, \tag{56}$$

where H_3 is the Hamiltonian for the sub-system previously described by Equation (7) The perturbing function $\mathscr{R}$ is then defined by

$$\mathscr{R} = \tfrac{1}{2}M\mathbf{v}_0^2 + \tfrac{1}{2}\sum_{i=4}^{N} m_i\mathbf{v}_i^2 - \tfrac{1}{2}\sum_{i=4}^{N}\sum_{\substack{j=4\\ j\neq i}}^{N}\frac{m_i m_j}{|\mathbf{r}_i - \mathbf{r}_j|} - \sum_{i=1}^{3}\sum_{j=4}^{N}\frac{m_i m_j}{|\mathbf{r}_i - \mathbf{r}_j|}, \tag{57}$$

where $M = m_1 + m_2 + m_3$ and $\mathbf{v}_0$ is the associated centre of mass velocity,

$$\mathbf{v}_0 = \frac{1}{M}\sum_{i=1}^{3} m_i\mathbf{v}_i . \tag{58}$$

Regularization of the local Hamiltonian H_3 follows the development of the preceding sections and the formulation in the extended phase space is preserved by adopting the constraint $P_9 = -H_3$ along the solution path. Substitution for H_3 from Equation (56) finally results in the modified Hamiltonian function

$$\Gamma^* = R_1R_2(\mathscr{R} - H_N) + R_1R_2\tilde{H}_3, \tag{59}$$

where the expression for $\tilde{H}_3$ is given explicitly by Equation (29). In the following it is convenient to write the equations of motion in vector form. We also note that $\mathscr{R}$ does not depend on the momenta $\mathbf{P}_k$, hence the new equations of motion become

$$\begin{aligned}
\frac{\mathrm{d}\mathbf{Q}_k}{\mathrm{d}\tau} &= \frac{\partial}{\partial\mathbf{P}_k}(R_1R_2\tilde{H}_3), \qquad (k = 1, 2)\\
\frac{\mathrm{d}\mathbf{P}_k}{\mathrm{d}\tau} &= -\frac{\partial}{\partial\mathbf{Q}_k}(R_1R_2\tilde{H}_3) - (\mathscr{R} - H_N)\frac{\partial}{\partial\mathbf{Q}_k}(R_1R_2) - R_1R_2\frac{\partial\mathscr{R}}{\partial\mathbf{Q}_k}.\\
&\qquad\qquad (k = 1, 2)
\end{aligned} \tag{60}$$

This set of equations is analogous to the two-body formulation of Peters (1968) and is also formally similar to the classical variation of parameters method, where only the equations for the velocities are modified by the perturbation. The second of Equations (33) no longer applies. Because of the form of Equation (56) with H_N being a constant of the motion, it should be replaced by the perturbation equation

$$\frac{\mathrm{d}P_9}{\mathrm{d}\tau} = \frac{\mathrm{d}\mathscr{R}}{\mathrm{d}\tau}. \tag{61}$$

Alternatively, we note that the energy relation (56) implies that the term $\mathscr{R} - H_N$ required by the second of Equations (60) is equivalent to $-H_3$. If the particle number is large, the time-consuming effort of evaluating the former expression may therefore be avoided by integrating the perturbation equation $\mathrm{d}H_3/\mathrm{d}\tau$ directly. It can be readily verified that this approach leads to a differential equation which is of the same form as Equation (61).

The perturbing function (57) depends on the physical variables $\mathbf{r}$ and $\mathbf{v}$, whereas Equation (60) introduces the variation of $\mathscr{R}$ with respect to regularized quantities.

Accordingly, we first write the relevant global coordinates in terms of relative distances which may then be differentiated. The first and second of Equations (55), together with the centre of mass condition

$$\mathbf{r}_0 = \frac{1}{M} \sum_{i=1}^{3} m_i \mathbf{r}_i , \tag{62}$$

give rise to the respective coordinates

$$\begin{aligned} \mathbf{r}_1 &= \mathbf{r}_0 + (m_2 + m_3)\, \mathbf{q}_1/M - m_2 \mathbf{q}_2/M , \\ \mathbf{r}_2 &= \mathbf{r}_0 - m_1 \mathbf{q}_1/M + (m_1 + m_3)\, \mathbf{q}_2/M , \\ \mathbf{r}_3 &= \mathbf{r}_0 - m_1 \mathbf{q}_1/M - m_2 \mathbf{q}_2/M . \end{aligned} \tag{63}$$

The desired perturbation term may now be obtained from

$$\frac{\partial \mathscr{R}}{\partial \mathbf{Q}_k} = \sum_{i=1}^{3} \frac{\partial \mathscr{R}}{\partial \mathbf{r}_i} \frac{\partial \mathbf{r}_i}{\partial \mathbf{q}_k} \frac{\partial \mathbf{q}_k}{\partial \mathbf{Q}_k} , \qquad (k = 1, 2) \tag{64}$$

since the terms $\partial \mathscr{R}/\partial \mathbf{v}_i$ vanish for $i = 1, 2, 3$. Explicit differentiation of Equation (57) yields the perturbation in Cartesian coordinates,

$$\frac{\partial \mathscr{R}}{\partial \mathbf{r}_i} = - m_i \mathbf{F}_i , \tag{65}$$

where $\mathbf{F}_i$ is the perturbing acceleration defined by

$$\mathbf{F}_i = - \sum_{j=4}^{N} m_j \frac{\mathbf{r}_i - \mathbf{r}_j}{|\mathbf{r}_i - \mathbf{r}_j|^3} . \qquad (i = 1, 2, 3) \tag{66}$$

The expressions for $\partial \mathbf{r}_i / \partial \mathbf{q}_k$ reduce to simple mass ratios by Equations (63), whereas differentiation of the transformations (40) may be written in compact notation as

$$\frac{\partial \mathbf{q}_k}{\partial \mathbf{Q}_k} = \mathbf{A}_k . \qquad (k = 1, 2) \tag{67}$$

Combining terms, Equation (64) simplifies to

$$\frac{\partial \mathscr{R}}{\partial \mathbf{Q}_k} = - \frac{\mathbf{A}_k}{M} \left[m_1 m_2 (\mathbf{F}_2 - \mathbf{F}_1)\, (-1)^k + m_k m_3 (\mathbf{F}_k - \mathbf{F}_3) \right] . \qquad (k = 1, 2) \tag{68}$$

This result may also be used to evaluate the right-hand side of Equation (61) because

$$\frac{\mathrm{d} \mathscr{R}}{\mathrm{d} \tau} = \sum_{k=1}^{2} \frac{\partial \mathscr{R}}{\partial \mathbf{Q}_k} \frac{\mathrm{d} \mathbf{Q}_k}{\mathrm{d} \tau} . \tag{69}$$

Substitution for the Hamiltonian in the first of Equations (60) gives the equation of

motion

$$\frac{d\mathbf{Q}_k}{d\tau} = \frac{1}{4\mu_{k3}} R_l \mathbf{P}_k + \frac{1}{16 m_3} \mathbf{A}_k \mathbf{A}_l^T \mathbf{P}_l, \qquad (k = 1, 2) \tag{70}$$

where l is the integer part of $(k+1)/k$. Combining expressions by making use of the orthogonality condition (25), we finally obtain after some simplification

$$\frac{d\mathscr{R}}{d\tau} = -\tfrac{1}{4} \sum_{k=1}^{2} R_l \mathbf{P}_k^T \mathbf{A}_k (\mathbf{F}_k - \mathbf{F}_3). \tag{71}$$

It is apparent that Equation (71) is a generalization of the corresponding expression for perturbed two-body motion (Aarseth, 1971). Alternatively, this result may be derived from the expression

$$\frac{d\mathscr{R}}{d\tau} = R_1 R_2 \sum_{k=1}^{2} \sum_{i=1}^{3} \frac{\partial \mathscr{R}}{\partial \mathbf{r}_i} \frac{\partial \mathbf{r}_i}{\partial \mathbf{q}_k} \frac{d\mathbf{q}_k}{dt}, \tag{72}$$

where the equation of motion for $\mathbf{q}_k$ is given by the first of Equations (14). Explicit differentiation yields

$$\frac{d\mathbf{q}_k}{dt} = \frac{1}{\mu_{k3}} \mathbf{p}_k + \frac{1}{m_3} \mathbf{p}_l, \qquad (k = 1, 2) \tag{73}$$

whereupon substitution for $\mathbf{p}_k$ by Equation (26), combined with Equations (63) and (65) leads to the result established above.

Complete solutions of the perturbed three-body motions may be obtained by integrating the corresponding centre of mass as well as the $N-3$ other particles. The appropriate equations of motion are

$$\frac{d^2\mathbf{r}_0}{dt^2} = \frac{1}{M} \sum_{i=1}^{3} m_i \mathbf{F}_i,$$

$$\frac{d^2\mathbf{r}_i}{dt^2} = -\sum_{\substack{j=1 \\ j \neq i}}^{N} m_j \frac{\mathbf{r}_i - \mathbf{r}_j}{|\mathbf{r}_i - \mathbf{r}_j|^3}. \qquad (i = 4, 5, \ldots, N) \tag{74}$$

If desired, these equations may be split into an equivalent set of first-order equations and expressed with respect to the regularized time by multiplying the new right-hand sides by $R_1 R_2$. However, the strategy to be adopted depends on the integration method itself as well as the problem under consideration. In either case, the perturbations $\mathbf{F}_i$ must be calculated in Cartesian coordinates, using the transformations (63).

The unperturbed formulation makes use of the centre of mass integrals to reduce the system of first-order equations from 18 to 12. This procedure is analogous to the introduction of Jacobian coordinates for describing the two-body problem. Since the centre of mass motion is integrated, it follows that more than one triple system may

be treated at the same time. A suitable computational procedure may be generalized from the corresponding treatment of simultaneous two-body encounters (Aarseth, 1971) and need therefore not be discussed here.

7. Numerical Tests

A variety of examples have been studied in order to assess the effectiveness of the new method. In the first instance, it is instructive to make comparisons with the standard KS regularization as applied to the three-body problem. The method of solution adopted by Bettis and Szebehely (1971) treats all equations of motion in regularized time and appears to be the most accurate version available. Both numerical procedures employ a Runge-Kutta 7(8)-order integrator with automatic stepsize control (Fehlberg, 1968) and the comparison results were obtained on a CDC 6600 computer which has an accuracy of about 14 significant figures.

TABLE I

Initial conditions for comparison problems

Example	m_i	x_i	y_i	dx_i/dt	dy_i/dt
	3.0	1.0	3.0	0	0
I	4.0	−2.0	−1.0	0	0
	5.0	1.0	−1.0	0	0
	1.0	0	0.002	−11.18	0
II	1.0	0	−0.002	11.18	0
	1.0	0.02	0	−22.36	0

Initial conditions for two test problems are given in Table I. Only motions in the xy-plane are considered here but the numerical integrations are performed by the general procedure. Example I is the much studied 'Pythagorean' problem (Szebehely and Peters, 1967) which poses a severe test of the ability to deal with close two-body encounters. The second example leads to a triple encounter which is not too critical to be attempted by the single KS transformation method. In this test, one of the particles is approaching a binary of zero eccentricity along a hyperbolic orbit initially aimed at the binary centre of mass. Because of the strong interaction, the incoming particle is captured, whereas the second particle escapes and leaves behind a wider binary. This type of motion occurs frequently in N-body calculations and leads to considerable loss of accuracy in all current methods. It is also of interest to consider critical triple encounters for systems with negative total energy and some examples will be discussed subsequently.

The parameters summarized in Table II form the basis of the comparison. The integration of example I was terminated at time $t_f = 16.0$. During this time the second and third particles experience four encounters within a distance of 0.01, whereas the closest approach between the first and third particles is about 0.1. On the other hand,

TABLE II

Accuracy and timing parameters

Method	Example	Tolerance	$\lvert\Delta H'/H'\rvert$	Number of steps [a]	Seconds
Standard KS	I	10^{-8}	3×10^{-9}	132	2.32
Equation (44)	I	10^{-8}	2×10^{-11}	129	1.81
Standard KS	II	10^{-8}	2×10^{-10}	73	1.51
Equation (44)	II	10^{-8}	4×10^{-9}	38	0.60
Equation (44)	II	10^{-9}	4×10^{-10}	50	0.75
Equation (44)	II	10^{-11}	2×10^{-12}	87	1.22

[a] Not including the final time iteration.

the smallest distance between the first and second particles is about two length units. Consequently, the system experiences a rather weak triple encounter and there is no need to change the reference body. Example II is more realistic in the sense that a re-labelling is required three times before the termination at $t_f = 0.002$. It is also more difficult to maintain the accuracy of the solutions since the minimum triple configuration is considerably smaller than the initial value; i.e. $\min(R) \lesssim 2 \times 10^{-3}$.

The numerical procedure controls the integration accuracy by restricting the rms truncation error according to the tolerance. However, the equations of motion take different forms in the two regularizations and the overall accuracy measured by energy conservation is therefore not uniquely related to the tolerance. For this reason, the relative error of total energy given in column four should be considered in conjunction with the corresponding number of integration steps displayed in column five. When judged by the energy errors of example I, the new method is more accurate for the same number of integration steps. Conversely, the results of example II demonstrate that a similar accuracy may be obtained with a reduced number of function calls. Since a small energy error does not by itself guarantee the accuracy of the solutions, a detailed comparison has been made with respect to the 10^{-11} tolerance results. The rms deviation of the individual coordinates is about 3×10^{-12} for both methods when the respective number of steps is 73 and 50, hence in this example the lack of energy conservation is also a good measure of the integration errors.

Although accuracy considerations are of primary importance when discussing different methods, the results must be judged in relation to the amount of calculations involved. For comparison, the last column of Table II contains the computing time in seconds. It is expected that the new method would also be competitive in speed if the centre of mass integration were included, as required by the corresponding perturbed formulation.

In order to establish the practical limitations of the new method, it is desirable to investigate critical interactions near the triple collision singularity. We adopt the simple procedure of generating arbitrarily close encounters by perturbing the coordinates of an equilateral triangular configuration in which the three particles of mass

unity are initially at rest. Let the initial xy-coordinates before application of the centre of mass correction be $(-1, 0)$, $(1, 0)$ and $(\delta x, \sqrt{3}+\delta y)$, where δx and δy denote the departures from perfect symmetry. If $\delta x=0$, the exact solution path of one of the particles should lie on the y-axis and subsequent displacements may therefore be used as an indication of the integration accuracy. However, in the following we rely primarily on the more stringent test involving time reversal of the equations of motion.

TABLE III

Critical triple encounters

δx	δy	Number of steps	$\frac{1}{3}C$	R_f/a_f	$[\frac{1}{3}\sum_{i=1}^{3}(\Delta x_i^2+\Delta y_i^2)]^{1/2}$
0	1×10^{-4}	250	4×10^{-4}	20	1×10^{-9}
0	1×10^{-6}	250	8×10^{-6}	18	2×10^{-8}
0	1×10^{-8}	250	1×10^{-7}	13	9×10^{-8}
1×10^{-4}	0	250	3×10^{-4}	18	2×10^{-10}
1×10^{-6}	0	250	6×10^{-6}	15	2×10^{-9}
0	0	200	1×10^{-12}	–	1×10^{-6}

The parameters of six representative examples are displayed in Table III. In the first three examples, the perturbed particle escapes along the negative y-axis and leaves behind a close binary, whereas the next two examples also lead to binary formation, with the particle initially at $(-1, 0)$ escaping in the first quadrant. These results were obtained using 16 figure precision of an IBM 360/44 computer and a tolerance of 10^{-12}. The forward calculations were terminated after a fixed number of integration steps shown in column three. In view of the numerical error growth, any time reversal test may be expected to have a limited applicability. It is particularly difficult to recover the initial conditions after a large number of binary revolutions subsequent to the critical interaction, because the numerical solutions exhibit an instability which grows in both the forward and backward integration. Close binaries form in all the first five examples, with an osculating semi-major axis nearly equal to one third of the minimum triple perimeter $C=R_1+R_2+R$, displayed in column four. The calculations were continued beyond the critical stage as shown by column five which gives the final distance of the escaper in terms of the semi-major axis. The last column contains the rms deviation of the coordinates, calculated at the end of the time reversal. Inspection of the actual results reveals that these errors would be consistently smaller if the time reversed solutions were terminated slightly earlier, indicating that the treatment of the time leads to greater systematic errors than those in the orbits themselves.

The particle with initial coordinates $(0, \sqrt{3}+\delta y)$ remains on the y-axis as long as it is associated with m_3, whereas the accuracy of the corresponding y-coordinate is not maintained. In any case, the symmetrical description cannot be imposed if the other two particles collide and some displacement from the y-axis is therefore inevitable. In the calculations quoted so far, the numerically induced asymmetry is suffi-

ciently small ($<10^{-12}$) during the collapse stage that the results are not significantly affected.

Apart from integration errors which are common to any numerical procedure, the new method introduces an additional uncertainty whenever a re-labelling is performed. The latter contribution is not important, unless transformations from regularized to physical variables occur during close encounters. However, by an appropriate initial choice of particle labels, many triple encounters may also be studied without permitting re-labelling during the critical stage. The examples of Table III are of this type and these results were therefore obtained with fixed particle labels in order to evaluate the numerical performance by itself. For comparison, use of the standard switching condition $R > \min(R_1, R_2)$ involves two transformations in the first five examples and leads to a deterioration of the time-reversed coordinate deviations by factors in the range 10–10^5.

Equilateral three-body configurations, started from rest, should lead to a triple collision irrespective of the choice of masses. Although the numerical calculations do not preserve the symmetry exactly, it is challenging to test the time reversibility of such systems. Once introduced, departures from symmetry continue to grow by dynamical instabilities until the collapse is inevitably halted, resulting in binary formation with one of the particles escaping rapidly. The approximate minimum configuration for a tolerance of 10^{-12} is shown in the last entry of Table III, together with the rms deviation of the time-reversed solution, obtained before the distance R becomes significantly smaller than R_1 and R_2. Additional tests with slightly different initial conditions ($\delta y = 10^{-14}$) show essentially the same behaviour, indicating that the numerical errors are mainly responsible for the final asymmetry. In any case, the inability to specify $\sqrt{3}$ exactly would eventually impose its own lower limit on the contraction.

8. Discussion

At first sight it may seem surprising that the new method cannot be used to study three-body systems if one of the particles is massless. The failure to deal with the restricted problem is not due to the momentum formulation itself, since the mass singularities associated with the kinetic energy terms may be removed by using regularized velocities. At the same time, the new method is closely related to the standard KS regularization which permits one of the two particles to be massless.

In order to clarify these remarks, we first consider the analogous Hamiltonian for the isolated system of particles m_1, m_3 (cf. Figure 1) by omitting all terms connected with m_2 from Equation (44), including the factor R_2. This simplification leads to a Hamiltonian which is identical to the unperturbed KS formulation of Peters (1968). The mass singularity associated with $1/\mu_{13}$ may now be made a common factor by introducing the regularized velocity $\mathbf{V}_1 = \mathbf{P}_1/\mu_{13}$ as well as the binding energy per unit mass, $h = -P_9/\mu_{13}$. Finally, the reduced mass cancels on both sides when writing the second equation of motion (36) in terms of the variable $\mathbf{V}_1$, whereas the first of Equations (36) implies that $\mathbf{V}_1 = 4\,\mathrm{d}\mathbf{Q}_1/\mathrm{d}\tau$.

Returning now to the restricted three-body problem with $m_2=0$, the same treatment fails to make m_2 a common factor of all terms involved in the differentiation with respect to $\mathbf{Q}_2$. The special case of a massless reference body is equally critical, since the Hamiltonian appears to contain a non-removable singularity associated with m_3, which can only be eliminated by introducing additional equations. Moreover, these difficulties indicate that the numerical results may be less reliable if the mass ratios are very large.

The present regularization formulation is based on well-known dynamical concepts combined with two KS transformations. It may now be seen that the initial step of reducing the order of the differential equations is merely a convenient device for describing the internal motions, in complete analogy with the derivation of unperturbed two-body regularization. However, a practical solution would not have been possible without the simple form of the generating function (17) which permits the relative motions to be treated as two coupled Keplerian systems. The corresponding singularities in the equations of motion are then reduced to the same type by means of two simultaneous KS transformations and are finally removed by standard techniques.

The time transformation (43) regularizes all terms in the Hamiltonian associated with the distances R_1 and R_2, leaving one singularity due to the potential energy term m_1m_2/R. After differentiation, the equations of motion for the momenta $\mathbf{P}_k$ contain two singular terms of the type $\mathbf{Q}_kR_l/R$ and $R_1R_2\mathbf{A}_k\mathbf{R}/R^3$, where l is the integer part of $(k+1)/k$. From the geometry of the problem, $R\simeq R_1$ if $R>R_2$ and $R_1>R_2$, hence in most cases the inequality $R>\min(R_1, R_2)$ ensures that the critical terms are numerically smaller than the regularized contributions. This desirable property compensates for the divisions and is mainly responsible for the improved treatment of close triple encounters, whereas binary collisions present no problems. Implementation of the switching procedure requires a transformation to physical variables before assigning new particle labels, whereupon corresponding regularized variables are introduced and the integration may proceed. However, when the masses are different, the heaviest particle tends to dominate the motions and is then a natural reference body, thereby reducing the number of switching transformations.

Apart from the improved accuracy, the new method has considerable practical advantages. First, the actual decision-making and frequency of re-labelling is reduced significantly with respect to earlier procedures. Moreover, the equations of motion are of a simple type where all coordinates and momenta are treated similarly. In addition to studies of three-body systems, it is anticipated that applications of the perturbed formulation will lead to further improvements of N-body calculations where critical triple encounters have proved troublesome for two-body regularization methods. On the other hand, it may still be more appropriate to use the latter in order to study close binaries without a natural neighbour. Implementation of the full computational procedure is similar in all these methods, except that each sub-system of three particles is now described by 24 equations of motion as compared to 22 equations in the corresponding standard KS treatment. The total number of force calculations is unchanged, however, and the computational effort per integration step is therefore similar.

As in the single KS transformation method, several sub-systems may be regularized simultaneously, using either a common time transformation (Bettis and Szebehely, 1971) or treating each configuration separately (Aarseth, 1971). Although the former procedure is more accurate in the sense that the motions of other particles are regular in the presence of one binary collision, it is less efficient for studying large particle numbers. Furthermore, the problem of treating two simultaneous binary collisions is closely related to the problem of triple collisions, since in both cases the time transformation is of the type $\mathrm{d}t = R_1 R_2 \, \mathrm{d}\tau$ which leads to an improper integral if both distances tend to zero together. This behaviour is of no consequence in practical calculations, as shown by the critical triple encounters studied here.

In view of the fundamental difficulties associated with multiple collisions, regularization methods are mainly concerned with the removal of singularities between pairs of particles. This approach is especially effective in the study of close two-body encounters, provided that the external perturbation is not too large. The concept of dominant two-body motion also plays a significant role in the present method, since the proper reference body minimizes the interaction term $m_1 m_2 / R$. Hence if the masses differ significantly, the geometrical re-labelling condition employed here should be replaced by an appropriate dynamical criterion.

In conclusion, we note that the present regularization method may be generalized to deal with an arbitrary number of particles. The required development is relatively straightforward and consists of introducing additional KS transformations for the interactions between the other particles and the original reference body. In the general case, the members of an arbitrary sub-system may be regularized in a similar way, leaving the remaining particles to be treated by time smoothing (Zare, 1974). However, it is unlikely that more than two particles would normally share the same natural reference body. Because of the inverse cube singularity in the equations of motion, it would then no longer be possible to minimize all interaction terms. The generalized formulation may therefore be practically useful only in special systems containing one dominant member. Alternatively, if several distances become small independently of each other, a compact configuration originally treated as one system may be subdivided. This difficulty is not present in the new global regularization of Heggie (1974) but it remains to be seen which procedure is best suited to actual N-body calculations.

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